

VULNER

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Bash-based Automation Tool for Network Service & Vulnerability Mapping

Penetration Testing Project - ZX301



Introduction

VULNER is a Bash-based penetration testing automation tool designed for authorized private lab networks. It automates service discovery, TCP and UDP scanning, weak credential assessment, vulnerability mapping, result parsing, reporting, and evidence packaging.

Key Capabilities

- Validates private IPv4 targets
- Runs TCP and UDP service/version scans
- Parses discovered hosts and services
- Detects login services for controlled credential testing
- Supports Hydra/Medusa-based weak credential assessment
- Generates summaries, HTML/Markdown reports, and ZIP evidence archives

Why VULNER Was Built

VULNER was developed to simplify and automate repetitive penetration testing tasks in an authorized private lab environment. Instead of manually switching between multiple tools, commands, output files, and result formats, the script brings the main workflow into one structured process.

The Challenge

- Manual scanning requires switching between tools and command formats
- TCP, UDP, credential testing, and vulnerability mapping produce separate outputs
- Important findings can be missed when results are not organized
- Brute-force testing must be controlled and limited to authorized targets
- Final evidence must be collected clearly for review and submission

The Solution

- VULNER automates TCP and UDP service discovery
- It validates private IPv4 targets before scanning
- It parses discovered hosts, services, and login ports
- It supports controlled Hydra/Medusa weak credential testing
- It maps potential vulnerabilities using Nmap NSE and Searchsploit
- It generates summaries, reports, logs, and a ZIP evidence archive

Initial Script Execution

The script begins by displaying the VULNER banner and asking the user to define the authorized assessment scope.

Execution Highlights

- Clear startup banner identifies the tool
- Target scope is configured before any scan
- User input is required before execution continues
- The workflow begins with an authorized assessment boundary

```
(kali㉿kali)-[~/scripts]
$ ./TMagen773640.s11.ZX301.sh

VULNER
Network Service and Vulnerability Mapper

Target Scope Setup

Enter private network target scope (single IP or CIDR):
```

Target Scope Setup

At the beginning of the assessment, the user defines the private target scope and selects the output directory. VULNER then prepares a dedicated results folder for the current run.

Setup Highlights

- The target can be entered as a single IP address or a CIDR range
- Only private network targets are accepted
- The output directory is defined before the scan begins
- VULNER automatically creates a dedicated results folder for the assessment

Target Scope Setup

```
Enter private network target scope (single IP or CIDR): 192.168.72.0/24
Enter output directory [./results]:
[OK] Output directory prepared: ./results/vulner_20260601_143524
```

Environment and Tool Check

Before running any scan, VULNER checks whether the required tools are installed and available in the system path. This ensures that the assessment can continue without missing dependencies.

Validated Tools

- Nmap is used for TCP, UDP, and service/version scanning
- Hydra and Medusa are used for controlled credential testing
- Searchsploit is used for vulnerability research and exploit mapping
- Zip is used to package the final evidence archive
- Grep, Awk, and Sed are used for parsing and result processing

[1/9] Environment and tool check

```
[OK] nmap: /usr/bin/nmap
[OK] nmap is installed.
[OK] hydra: /usr/bin/hydra
[OK] hydra is installed.
[OK] medusa: /usr/bin/medusa
[OK] medusa is installed.
[OK] searchsploit: /usr/bin/searchsploit
[OK] searchsploit is installed.
[OK] zip: /usr/bin/zip
[OK] zip is installed.
[OK] grep: /usr/bin/grep
[OK] grep is installed.
[OK] awk: /usr/bin/awk
[OK] awk is installed.
[OK] sed: /usr/bin/sed
[OK] sed is installed.
```

Input Validation and Scan Mode Selection

After validating the private target scope, VULNER asks for confirmation before scanning the network. The user then selects the scan mode: Basic for service discovery and credential testing, or Full for additional NSE vulnerability scripts and Searchsploit mapping.

Validation Highlights

- Confirms the accepted private network scope
- Requires user approval before scanning a CIDR range
- Offers Basic and Full scan modes
- Full mode adds NSE vulnerability scripts and Searchsploit mapping

[2/9] Input and scope validation

```
[OK] Target scope accepted: 192.168.72.0/24
[OK] Output directory: ./results/vulner_20260601_143524
[OK] Target scope validated.
[INFO] Network scan target: 192.168.72.0/24
Continue? [y/N]: y
[OK] Network scan confirmed.
```

Choose scan mode:

- 1) Basic - TCP/UDP scans, service/version detection, and controlled weak-password testing
- 2) Full - Basic plus NSE vulnerability scripts and Searchsploit mapping

Selection [1]: ☐

Basic Scan Mode

In Basic mode, VULNER loads username and password lists for controlled weak credential testing. The user can provide custom wordlist paths or continue with the default wordlists.

Basic Mode Setup

- Basic scan mode was selected
- Username and password wordlists were provided
- Wordlist entries were counted and displayed
- Credential testing remains controlled and optional

```
Choose scan mode:
```

- ```
 1) Basic - TCP/UDP scans, service/version detection, and controlled weak-password testing
 2) Full - Basic plus NSE vulnerability scripts and Searchsploit mapping
```

```
Selection [1]: 1
```

```
[OK] Scan mode selected: Basic
```

```
Username list path [/usr/share/seclists/Usernames/top-usernames-shortlist.txt]:
```

```
Password list path [/usr/share/wordlists/rockyou.txt]: /home/kali/scripts/rockyou_top200.txt
```

```
[OK] Username list: /usr/share/seclists/Usernames/top-usernames-shortlist.txt
```

```
[INFO] Username entries: 17
```

```
[OK] Password list: /home/kali/scripts/rockyou_top200.txt
```

```
[INFO] Password entries: 200
```



# Basic Nmap Scan Commands

In Basic mode, VULNER runs two Nmap scans: a TCP service/version scan and a UDP top-ports scan. Both commands save their results for later parsing, analysis, and reporting.

## Scan Breakdown

TCP:

-sS: TCP SYN scan

-sV: Service/version detection

Shared Options:

-T3: Balanced scan timing

--reason: Shows detection reasoning

-oA: Saves results in normal, grepable, and XML formats

UDP:

-sU: UDP scan

--top-ports 20: Scans the most common UDP ports

### [4/9] TCP service/version scan

```
[INFO] Running TCP service/version scan
[INFO] Command: nmap -sS -sV -T3 --reason -oA ./results/vulner_20260601_143524/nmap/tcp_services 192.168.72.0/24
[INFO] Running TCP service/version scan finished with exit code 0
```

### [5/9] UDP top ports scan

```
[INFO] Running UDP top ports scan
[INFO] Command: nmap -sU --top-ports 20 -T3 --reason -oA ./results/vulner_20260601_143524/nmap/udp_top 192.168.72.0/24
[INFO] Running UDP top ports scan finished with exit code 0
```

# Service Parsing

After the TCP and UDP scans are completed, VULNER parses the Nmap results and converts the raw scan output into a structured service summary. This stage displays the discovered hosts, ports, protocols, services, and available version information.

## Parsing Highlights

- Extracts open TCP and UDP services from the scan results
- Displays host, port, protocol, service name, and version details
- Summarizes the number of discovered services across multiple hosts
- Prepares structured output for reporting and later workflow stages

### [6/9] Service parsing

[OK] Parsed 33 open service entries across 4 host(s).

#### Discovered Services

| Host           | Port | Protocol | Service     | Version                                      |
|----------------|------|----------|-------------|----------------------------------------------|
| 192.168.72.129 | 21   | tcp      | ftp         | vsftpd 2.3.4                                 |
| 192.168.72.129 | 22   | tcp      | ssh         | OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0) |
| 192.168.72.129 | 23   | tcp      | telnet      | Linux telnetd                                |
| 192.168.72.129 | 25   | tcp      | smtp        | Postfix smtpd                                |
| 192.168.72.129 | 53   | tcp      | domain      | ISC BIND 9.4.2                               |
| 192.168.72.129 | 80   | tcp      | http        | Apache httpd 2.2.8 ((Ubuntu) DAV 2)          |
| 192.168.72.129 | 111  | tcp      | rpcbind     | 2 (RPC #100000)                              |
| 192.168.72.129 | 139  | tcp      | netbios-ssn | Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  |
| 192.168.72.129 | 445  | tcp      | netbios-ssn | Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  |
| 192.168.72.129 | 512  | tcp      | exec        | netkit-rsh rexecd                            |
| 192.168.72.129 | 513  | tcp      | login       | OpenBSD or Solaris rlogind                   |

# Controlled Weak Credential Testing

After detecting login services, VULNER allows the user to run controlled weak credential testing against a specific discovered host. The test is not executed automatically and requires user confirmation.

## Credential Testing Controls

- Detects login services such as SSH, FTP, Telnet, and RDP
- Displays available targets before testing
- Allows the user to select a specific host
- Infers basic host roles based on discovered services
- Requires confirmation before running credential testing

### [7/9] Controlled weak credential testing

[OK] Discovered 5 login service entries.

192.168.72.128 ssh 22 tcp

192.168.72.129 ftp 21 tcp

192.168.72.129 ssh 22 tcp

192.168.72.129 telnet 23 tcp

192.168.72.137 rdp 3389 tcp

Do you want to run controlled weak credential testing against a specific discovered host? [y/N]: y

Available login targets:

| # | Host           | Role                    | Login services |
|---|----------------|-------------------------|----------------|
| 1 | 192.168.72.128 | Linux host (inferred)   | ssh            |
| 2 | 192.168.72.129 | Linux host (inferred)   | ftp ssh telnet |
| 3 | 192.168.72.137 | Windows host (inferred) | rdp            |
| q | Skip           |                         |                |

Pick a host to test [q]: █



# Credential Test Execution

After the target host and service are selected, VULNER allows the user to choose a credential testing tool, displays the exact command, and requires explicit confirmation before execution. The result is then saved into a dedicated credential output file.

## Execution Controls

- The user selects a specific host and service
- The user chooses between Hydra and Medusa
- The exact command is displayed before execution
- Credential testing requires explicit confirmation
- Results are saved to a dedicated output file

```
Pick a host to test [q]: 2

Target: 192.168.72.129
Services:
 1) ftp 21/tcp
 2) ssh 22/tcp
 3) telnet 23/tcp
 q) cancel

Pick a service [q]: 2
Use hydra or medusa? [hydra]: medusa

Exact command: medusa -h 192.168.72.129 -U "/usr/share/seclists/Usernames/top-usernames-shortlist.txt" -P "/home/kali/scripts/rockyou_top200.txt" -M ssh -n 22
Run this credential test against this single discovered service? [y/N]: y
[INFO] Command: medusa -h 192.168.72.129 -U /usr/share/seclists/Usernames/top-usernames-shortlist.txt -P /home/kali/scripts/rockyou_top200.txt -M ssh -n 22
[OK] Credential test completed. Output: ./results/vulner_20260601_150400/credentials/credential_results/medusa_192.168.72.129_ssh_22.txt E
xit code: 0
```

# Full Mode Vulnerability Mapping

In Full mode, VULNER extends the basic scanning workflow by adding a dedicated vulnerability mapping stage. After the initial TCP, UDP, and parsing phases are completed, the script runs Nmap NSE vulnerability scripts and maps the results with Searchsploit.

## NSE Scan:

--script vuln: Runs Nmap NSE vulnerability scripts

## Searchsploit:

--nmap: Maps Nmap XML results to related Exploit-DB entries

## Output:

-oA: Saves normal, grepable, and XML formats

CVE Summary: Extracts potential CVE identifiers

Manual Review: Findings require validation

## **[8/9] Vulnerability mapping, Full mode only**

```
[INFO] Running Nmap NSE vulnerability scan
[INFO] Command: nmap -sV --script vuln -T3 --reason -oA ./results/vulner_20260601_154925/nmap/vuln_nse 192.168.72.0/24
[INFO] Running Nmap NSE vulnerability scan finished with exit code 0
[INFO] Command: searchsploit --nmap ./results/vulner_20260601_154925/nmap/tcp_services.xml
[OK] Vulnerability outputs created: ./results/vulner_20260601_154925/parsed/vulnerability_summary.txt
```

### Vulnerability summary

Generated: 2026-06-01 15:55:09

NSE output base: ./results/vulner\_20260601\_154925/nmap/vuln\_nse

NSE nmap output: ./results/vulner\_20260601\_154925/nmap/vuln\_nse.nmap

NSE XML output: ./results/vulner\_20260601\_154925/nmap/vuln\_nse.xml

Searchsploit output: ./results/vulner\_20260601\_154925/parsed/searchsploit\_mapping.txt

Searchsploit exit code: 0

CVE summary: ./results/vulner\_20260601\_154925/parsed/cve\_summary.txt

CVE count: 5

Potential findings require manual validation.

# Search Module

At the end of the workflow, VULNER allows the user to search across all saved results. This helps quickly locate relevant findings, services, IP addresses, credentials output, or vulnerability indicators inside the generated evidence folder.

## Search Highlights

- Allows searching across saved assessment results
- Helps locate services, IP addresses, and findings
- Provides feedback when a search term is too general
- Supports quick review of collected evidence

### [9/9] Summary, search, archive, and report

```
Search inside saved results now? [y/N]: y
Enter search term: ssh
[INFO] Searching results for: ssh
./results/vulner_20260601_154925/credentials/credential_results/hydra_192.168.72.129_ssh_22.txt:4:Service: ssh
./results/vulner_20260601_154925/credentials/credential_results/hydra_192.168.72.129_ssh_22.txt:6:Command: hydra -L "/usr/share/seclists/U
list.txt" -P "/home/kali/scripts/rockyou_top200.txt" -s 22 ssh://192.168.72.129
./results/vulner_20260601_154925/credentials/credential_results/hydra_192.168.72.129_ssh_22.txt:12:[DATA] attacking ssh://192.168.72.129:2
./results/vulner_20260601_154925/credentials/credential_testing_plan.txt:12:Host: 192.168.72.128 Service: ssh Port: 22 Protocol: tcp
./results/vulner_20260601_154925/credentials/credential_testing_plan.txt:13: Hydra: hydra -L "/usr/share/seclists/Usernames/top-usernames
li/scripts/rockyou_top200.txt" -s 22 ssh://192.168.72.128
```



# Final Packaging and Summary

At the end of the workflow, VULNER packages the collected results into a ZIP archive, generates report files, and presents a final assessment summary. This provides a structured overview of the scan results, credential testing activity, and detected vulnerability indicators.

## Final Output Highlights

- Packages the assessment results into a ZIP archive
- Generates Markdown and HTML report files
- Displays a final summary of discovered hosts and services
- Summarizes login services and CVE-related findings
- Provides a consolidated overview of the assessment run

**Potential findings still require manual validation before being treated as confirmed vulnerabilities.**

## Final Summary

### Final Assessment Summary

Generated: 2026-06-01 16:22:17  
Script: TMagen773640.s11.ZX301.sh  
Target scope: 192.168.72.0/24  
Scan mode: Full  
Output directory: ./results/vulner\_20260601\_154925

### Counts:

- Hosts discovered: 4
- TCP services: 28
- UDP services: 2
- Total services: 30
- Login services: 5
- CVE count: 5

# Conclusion

VULNER successfully automates the main stages of a penetration testing workflow inside an authorized private lab environment. The tool validates the target scope, performs TCP and UDP scanning, parses discovered services, supports controlled weak credential testing, maps potential vulnerabilities in Full mode, and packages the evidence into reports and a ZIP archive.

## Project Outcome

- Built a complete Bash-based automation workflow
- Combined scanning, parsing, credential testing, and reporting
- Added safety controls before sensitive actions
- Generated structured evidence for review and submission
- Demonstrated practical use of Nmap, Hydra, Medusa, and Searchsploit